

DHIA EDDINE JEDIDI

AI / Machine Learning / Data Science Engineer

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SUMMARY

AI/ML Engineer focused on taking models from experimentation to production, with hands-on work in computer vision, NLP, and API deployment. Experience includes benchmarking, fine-tuning, latency optimization, and operating Docker-based inference systems, plus data warehousing and forecasting projects. Engineer in Computer Science, Data Science track at ESPRIT; open to relocation.

EXPERIENCE

AI Engineering Intern — Octomiro (Client: Tunifries)

Jan 2026 – Jul 2026

OctoNorm — automated visual quality control for a fry production line

- Automated visual quality control for a fry production line, replacing ~1 hour of manual stoppage with 6–10 seconds per tray across 301 live production runs.
- Built and benchmarked production pipelines for grounded segmentation and defect detection, including 10 candidate pipelines on 3,300+ annotated instances and a best model result of 0.967 AUROC on a 252-image benchmark.
- Deployed the system as a Flask + React application in Docker on an always-on V100S GPU server, with warm-loaded models and a retraining loop through the operator dashboard.

AI / ML Intern — El Fouladh

Jul 2025 – Sep 2025

- Built an OCR + BERT document classification pipeline with 96% accuracy on the KHATT benchmark.
- Deployed the pipeline as a Flask REST API and automated routing, reducing processing time by 90%.

Freelance AI Developer — YOLOv8 Object Detection

Feb 2025 – May 2025

- Fine-tuned YOLOv8 with targeted data augmentation, improving mAP from 0.72 to 0.91.
- Reduced inference latency by 45% through ONNX export and optimization.

AI Intern — MPSI

Feb 2023 – Jun 2023

- Built a CNN-based defect detection system in TensorFlow/Keras with 94% accuracy.
- Integrated the system into the production inspection workflow.

PROJECTS

BI & Analytics Platform — Cosmetic Supply Chain

PostgreSQL · Talend · Python · Jupyter · Power BI · Flask · Angular

- Designed a data warehouse centralising supply-chain processes, improving data accessibility by 70%, and automated ETL pipelines in Talend, cutting manual processing time by 60%.
- Built ML/DL demand-forecasting models (85% predictive accuracy) and interactive Power BI dashboards with dynamic KPIs, improving reporting efficiency by 50%.

Video Emotion Recognition — 3D CNN + Autoencoder + LSTM + Attention

PyTorch · OpenCV · Flask · Streamlit

- Implemented a multi-stage neural pipeline on CREMA-D; deployed as a Flask inference API with a Streamlit real-time webcam interface.

Twitter Influence Graph Analysis

BERTweet · GNN · Graph-RAG · PyTorch Geometric

- Fine-tuned BERTweet for sentiment analysis (92% accuracy, 1.6M tweets); built an influence graph of 150K nodes and 500K edges, applying GNN and Graph-RAG for influencer detection.

SKILLS

Machine Learning: scikit-learn · XGBoost · Statistics · Regression · Classification · Forecasting · Feature Engineering · Model Evaluation · Hyperparameter Tuning

Deep Learning: PyTorch · TensorFlow · Keras · CNNs · LSTMs · Transformers · Computer Vision · NLP · Object Detection · Image Segmentation · Anomaly Detection · OCR

LLMs & Generative AI: Hugging Face Transformers · BERT · RAG · LLM Agents · Prompt Engineering · OpenAI API

Cloud, MLOps & Deployment: Azure AI Foundry · Docker · MLOps · Model Deployment · Flask · FastAPI · REST APIs · ONNX Runtime · CUDA

Languages & Tools: Python · SQL · Git · Linux · Jupyter

Data & BI: PostgreSQL · ETL · Data Warehousing · Power BI · Data Visualisation

EDUCATION

Engineer in Computer Science — Data Science Track, ESPRIT, Tunis

Jan 2023 – Expected 2026

BSc Applied Mathematics: Data Science, FST, University of Tunis El Manar

2020 – 2023

CERTIFICATIONS & LANGUAGES

NVIDIA: Fundamentals of Deep Learning · AI for Predictive Maintenance · Transformer-Based NLP Applications

IBM / AWS: Deep Learning with PyTorch · AWS Cloud Foundations

The Hashgraph Association: Hashgraph Developer Course (Nov 2025)

Languages: Arabic (Native) · French (B2) · English (B2)